

Demant

Audio Explorers 2026

Software Case

Extracting the Talker of Interest in a Multi-Talker Sound Scene

Jonathan Tybirk

BSc Artificial Intelligence and Data, 6th semester

Technical University of Denmark (DTU)

April 2026

1 The Case

The cocktail party effect is the ability of listeners to focus on a particular speaker in a noisy environment where many conversations happen simultaneously. For people with hearing loss, this can be particularly difficult. Hearing aids should be able to assist users by singling out the speaker of most importance to the listener.

For this case, I analyze an audio mixture of concurrent speakers as recorded by a hearing aid. I extract information about each talker and identify a Talker of Interest, which I isolate and enhance for easy comprehension.

The full case description is included in Appendix B.

2 Method Validation on Example Mixture

Before analyzing the unknown mixture, I validated my methods on `example_mixture.wav`, where the number of speakers, their language, and their position relative to the listener are all known. My first goal was to separate the speakers, as clean isolations would make the other goals of retrieving speaker information and transcribing their speech much easier.

2.1 Microphone Geometry and Beamforming

As the hearing aid is equipped with four microphones, beamforming seemed like a natural option for speaker separation. In beamforming, microphone signals are delayed and summed so that sound from a target direction adds up constructively while sound from other directions cancels out, which can be used to isolate the sound from a particular direction.

Scanning across all directions and measuring the output power at each reveals the active speakers as peaks, allowing beamforming to be used for source localisation as well, giving the direction of arrival of each talker.

Beamforming does however require knowing the distances between microphones, which was not provided in the case. But I did have the speaker locations in `example_mixture.wav`, allowing me to essentially reverse-beamform to calculate the microphone geometry from the time delays between microphone pairs.

LF-LR and RF-RR gave two independent intra-ear spacing estimates, while LF-RF and LR-RR gave two independent inter-ear distance estimates. I averaged each independent pair for my first estimate. For an independent second estimate, I optimised over microphone geometries to find the distances for which the predicted time delays best matched the known speaker positions from the case.

Dimension	Avg Pairwise	Joint fit
Inter-ear distance	22.6 cm	21.8 cm
Intra-ear spacing	8.9 cm	1.6 cm

Table 1: Microphone geometry estimates from `example_mixture.wav`

As is immediately clear from Table 1, the intra-ear spacing estimates disagreed wildly. Disambiguating sources in front from sources behind would become a recurring problem throughout this case.

I experimented with MVDR and DUET beamforming variations for isolating speakers, but neither produced satisfactory separation. I quickly concluded that any method relying directly on time delays was unlikely to perform well, given the small microphone distances, particularly the few centimeters separating the front and rear microphones on each ear.

2.2 Independent Component Analysis

I next went for Independent Component Analysis (ICA), which requires no geometry assumptions and instead unmixes a signal into n components that are optimally independent of each other. I experimented with different variants like AuxIVA and ILRMA, as well as adding a Wiener filter to clean up residual bleed, but standard FastMNMF2 ended up performing by far the best on `example_mixture.wav`. I used Bayesian optimisation on the hyperparameters to further minimise cross-signal dependence, and ended up with 4 essentially perfectly isolated clips of one speaker each with no audible bleed.

FastMNMF2 also estimates the DoA of each separated source, and for `example_mixture.wav` it correctly placed speakers at the four cardinal directions. Interestingly, the English speaker was placed directly behind the listener rather than in front, as specified in the case. This again highlights how difficult it is to distinguish front from rear with these hearing aids, where the front and rear microphones are simply too close together to reliably break the ambiguity.

3 Mixture Analysis

At this point, I felt that ICA with Bayesian optimisation was validated as the method to move forward with, and applied the same pipeline to `mixture.wav`. I also vibe-coded an interactive frontend to make comparison of the growing number of algorithms feasible.

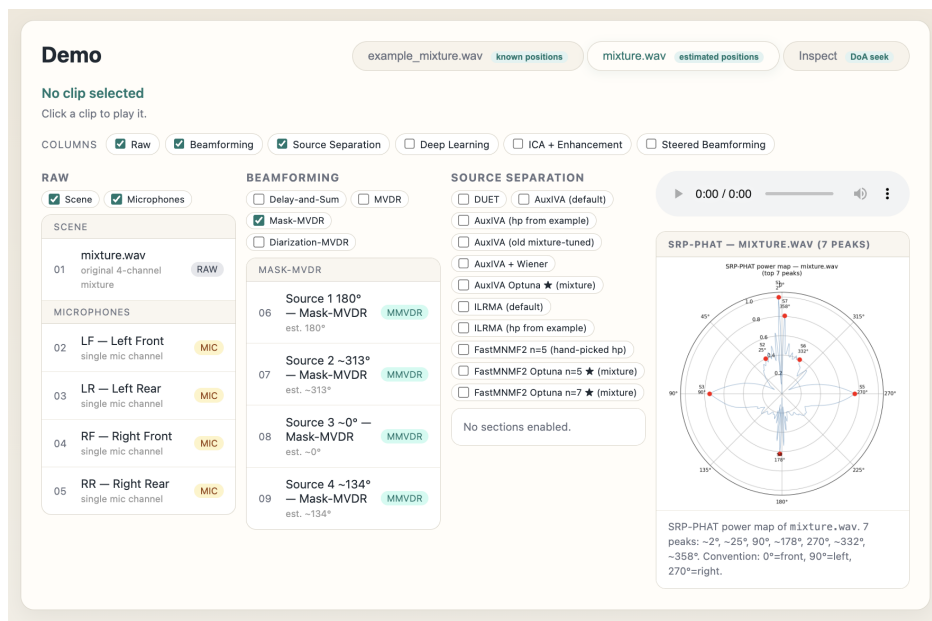


Figure 1: Interactive frontend used throughout the analysis to compare methods.

3.1 Speaker Identities

Running FastMNMF2 with $n=4$ on `mixture.wav` isolated groups of speakers well enough to start roughly transcribing them, revealing 7 unique speakers, two of which were in a conversation with each other. To keep track, I named them after salient topics in their speech: Australia Man, Birthday Brunch Woman, Mountain Man, Burning House Man, Ageing Man, Convo Man, and Convo Woman.

With 5 out of 7 people monologuing, I began to suspect that our Talker of Interest might be a partner in the Convo Couple, with us as the listener being the other. This is the only setup where I could see a case for any of the speakers being of particular interest to us above the others.

3.2 Initial Directions of Arrival

Five of the seven speakers appeared to be located roughly in front of or behind the listener according to both my Beamforming and ICA algorithms.

Luckily, the two speakers further to the side of us stood out immediately upon listening closely to the raw microphone channels. Australia Man came through markedly louder on the right microphones, while Birthday Brunch Woman was clearly audible on the left but almost inaudible on the right. This established their lateral positions without any algorithm. The remaining five speakers were too close to directly in front of or behind the listener to distinguish by ear, consistent with the front/rear limitation established earlier. This matched my experience listening to the raw microphone channels, as none of these five seemed to vary in volume between channels.

Convo Woman came through much clearer in the raw microphone channel than Convo Man, so I made the assumption that we are probably Convo Woman, with Convo Man being directly in front of us as the Talker of Interest.

3.3 Isolation of Talker of Interest

Unfortunately, running FastMNMF2 with $n=7$ was initially disappointing, grouping several speakers together per component and leaving the last few as near-pure noise. This makes sense given that 5 speakers are roughly in front of or behind the listener, making them particularly hard to separate.

Given that Australia Man and Birthday Brunch Woman were clearly to the sides of the listener, I could subtract them by discarding any part of the signal that was significantly louder on one side than the other. For each time-frequency bin, I computed the ratio of left to right microphone power and discarded bins where the imbalance exceeded 15%, leaving only bins corresponding to the 5 near-axis speakers.

Running FastMNMF2 on this masked signal isolated the Convo Couple, which I further post-processed with the `noisereduce` Python library to produce `target_speaker_enhanced.wav`.

I decided not to isolate Convo Man from Convo Woman, as hearing ourselves make the conversation easier to follow and as the two never speak at the same time anyway.

3.4 Final Directions of Arrival

With the Convo Couple isolated, I repeated the process for the remaining speakers, combining the same masking approach with FastMNMF2 runs at varying n to get individual isolations of each talker.

To get a more precise direction estimate for each speaker, I took their isolated audio, identified the moments where they were loudest, and ran a Beamforming scan on the raw mixture using only those moments. The two strongest peaks are reported as DoA#1 and DoA#2.

Speaker	DoA #1	DoA #2
Australia Man	97.5°	82.5°
Birthday Brunch Woman	258.5°	277.5°
Mountain Man	1.0°	0.0°
Convo Man	181.0°	0.0°
Convo Woman	180.5°	0.0°
Ageing Man	180.5°	97.5°
Burning House Man	258.5°	284.0°

Table 2: Per-speaker DoA estimates from activity-weighted direction scan.

The estimates agreed with the earlier listening-based conclusion that Australia Man is to the left and Birthday Brunch Woman to the right. For the near-axis speakers, the estimates are reasonably consistent with Convo Man coming from directly in front of the listener, which supports him as a plausible Talker of Interest.

4 Results

Table 3 summarises the identified speakers, their estimated directions of arrival, and their genders. Full transcriptions are provided in Appendix A.

Speaker	Gender	Est. DoA	Notes
Convo Man	M	~0° (front)	Talker of Interest
Convo Woman	F	~0° (front)	Wearer
Australia Man	M	~90° (left)	Confirmed
Birthday Brunch Woman	F	~270° (right)	Confirmed
Mountain Man	M	~0° or 180°	Front/rear ambiguous
Ageing Man	M	~0° or 180°	Front/rear ambiguous
Burning House Man	M	~0° or 180°	Front/rear ambiguous

Table 3: Identified speakers with estimated DoA and gender.

FastMNMF2 uses all four microphone channels simultaneously through spatial covariance modelling, making the separated output a weighted combination of all channels rather than any single microphone. This is preferable to picking one microphone, as it preserves spatial information and reduces noise without introducing directional artefacts.

The enhanced output is delivered as `talker_of_interest.wav`.

5 If I Had Had More Time

I was particularly interested in applying a source separation foundation model to this problem, but I was not able to find any competent model for this many speakers. I had planned to fine-tune models like SepFormer and TF-GridNet on a synthetic seven-speaker dataset built from LibriMix data. Unfortunately, I did not get approved for extra storage on my university’s high-performance computer in time to implement this before the deadline.

Other than fine-tuning a foundation model, I have not been able to come up with a good solution to the problem of distinguishing front from rear with microphones so physically close together. I am particularly curious to find out what other groups did here, if a reliable solution is even possible.

6 Towards Real Hearing Aids

Most of what I used here would not survive contact with a real hearing aid. FastMNMF2 with Bayesian optimisation takes 30 minutes to run offline on my university’s HPC cluster. The real-life hearing aid needs to process audio in real time on a low-power chip.

The lateral masking step is a possible exception. Comparing left and right ear loudness moment to moment is fast and makes no strong assumptions about the acoustic scene. The idea could plausibly generalise and run on hardware. That could help drown out noise coming from the sides of the person wearing the hearing aid, potentially letting them better focus on the person in front of them.

The front/rear problem is harder. It is a physical constraint of the device, and nothing I tried here came close to resolving it reliably. Beamforming and FastMNMF2 both showed the same ambiguity towards centered speakers. A well-trained deep learning model might learn to resolve it from spectral cues alone, which is part of why I was interested in fine-tuning SepFormer on matched data.

As long as the hearing aid is plugged into an outlet at all times + ethernet cable to a HPC cluster, it should run well on a real hearing aid too!

A Transcripts

Convo Couple

M: Hi, how are you?

W: Great, how are you?

M: I just had my birthday.

W: Really, how old are you?

M: I am 35.

W: Are you married?

M: Yes, I got married 3 years ago.

W: Do you have any children?

M: Yes, I have a daughter. She is 4 years old. Do you have children?

W: Yes, I have two. My older son is 5, and my younger son is 3.

Australia Man

Most sightseers will be Chinese, and for the great majority, this 83 million monument to life down under was the closest they'll come to visiting Australia. Australia's Commissioner-General to Expo, Lyndall Sachs, says: "There's a lack of knowledge here about Australia. They just think we blow stuff up out of the earth and export it."

Birthday Brunch Woman

Every year at my birthday brunch, guests elbow each other over the smoked salmon, while a bowl of pickled herring sits forlornly, mostly untouched. I should probably stop buying it, but I like it, and it's my birthday, after all. This year was no exception, and at the end—

Mountain Man

I had never thought about climbing mountain Kilimanjaro until I learned of its statistics. Over 25,000 climbers attack it each year, 40% can't make it to the top, and a handful don't make it down at all. A foolish curiosity set in. Might be a winner, or a loser.

Ageing Man

It is inevitable: the muscles weaken, hearing and vision fade, we get wrinkled and stooped, we can't eat, run or even walk as fast as we used to. We have aches and pains of our body never noticed before. We get old. It sounds miserable, but apparently, it's not. A large Gallup poll found that by almost any measure, —

Burning House Man

What would you save if your house was burning down? If a fire hit my house, I think I'd have this terrible guilt trip. I'd be running around picking up all the things that I haven't done. This teddy bear is not actually that special. I'd just feel really bad about having to exist. I've had it since I—

B Case Description

(See next pages)

Software Engineering Case: Extracting the talker of interest in a multi-talker sound scene

Audio Explorers, April 2026

Introduction and perspective

Communicating in multi-talker environments is challenging – even for individuals with normal hearing. For people with hearing loss, the challenge is significantly greater. In most cases, the difficulties extend beyond reduced audibility: the ability to discriminate between different sound sources is also impaired. As a result, an intelligible speech signal often requires reducing background noise in order to enhance the desired talker.

Identifying the desired talker is not trivial. Two listeners presented with the same sound mixture may still focus on different target talkers. Many hearing aids assume that the desired speaker is located in front of the listener. Alternatively, some systems select the most energetic talker. However, the most energetic source may differ between the left and right ears, leading to ambiguity.

The target talker may also change depending on the situation, e.g., a teacher in a classroom, colleagues at work, a partner or children at home, or any other familiar voice. During a phone call, even the user's own voice might be considered the target.

The more information we can extract from the sound scene, the better our chances of correctly selecting the talker of interest.

The case

Your goal is to extract as much information as possible from a provided multi-talker sound scene. The audio mixture contains several simultaneous talkers and is recorded using hearing-aid microphones mounted on a head, as illustrated in Figure 1. The audio is provided as a four channel WAV file with the channel order: **[left front, left rear, right front, right rear]**.

From this sound scene, you are asked to infer properties such as:

- The number of different talkers
- The direction of arrival (DoA) of each talker
- The gender of each talker
- A transcription of the individual talkers
- A reasoned guess regarding which talker is the talker of interest

You may also attempt to extract or enhance the talker of interest from the mixture.

In addition, an example recording from the same microphone array is provided. The corresponding positions of the four talkers are shown in Figure 2. These locations may be helpful when estimating the talker positions in the main sound scene.

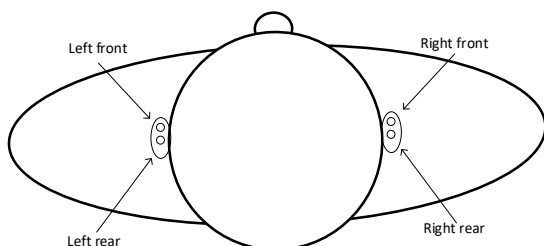


Figure 1:

Hearing aids mounted on the left and the right ear. A hearing aid typically contains two microphones. A front microphone and a rear microphone. An impinging sound from the front will arrive at the front microphone before the rear microphone and an impinging sound from the left will arrive at the left microphones before the right microphones. The time differences between the different microphones can be used to estimate the direction of arrival for each impinging sound source.

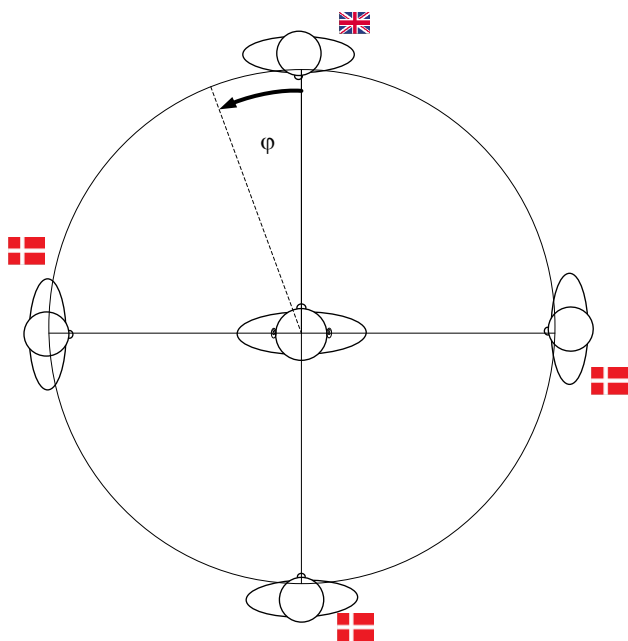


Figure 2: Example talker locations. The scene contains three Danish speaking and one English speaking talker. All talkers are approximately 1.9 meters from the listener and are positioned at azimuth angles of 0° , 90° , 180° , and 270° .

The solution

You may use any approach you prefer, including single microphone or multi microphone processing, machine learning models, or automatic speech transcription. Apply your chosen method to the provided audio mixture.

Delivery

You must write a **short report** documenting your process and reasoning.

The report must:

- Be no longer than **5 pages** (excluding the front page and optional appendices)
- Include each group member's name, affiliation, and current study semester
- Describe your approach, what worked and what did not, how you validated and tested your system, and your considerations regarding the microphone selection from the array

Once you are satisfied with your sound scene analysis, you may apply a target enhancement algorithm to the provided noisy mixture. Save the enhanced output as a **WAV file** and include it together with your report.

Ip assessment will be based primarily on the overall quality of your report, but the performance of your sound analysis system will also be considered.

Announcing the winners

The winning teams will be given direct notification via email or phone.

We hope you will find the challenge exciting and fun to work on, and we look forward to seeing your solutions.

Greetings,
The Audio Explorers Team